

Writing a GIMP 3.0 Plugin – Resources and Notes

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GIMP 3.0 is almost out, and I'm thrilled. Not only will this version bring new features and improvements, but it finally modernizes the technology stack in two important ways:

- It will use GTK 3
- It will use Python 3

This will make it easier and more sustainable to develop plugins for GIMP. The old Python 2 plugins will not work anymore, but the new Python 3 plugins will be more future-proof. I have a few ideas for plugins I want to write, so the end of last year I started to dig into it. And while I'm far from the only one with this idea, information on writing GIMP 3 plugins is naturally still sparse. So I decided to write down some collected resources and notes. If you have more resources or tips, feel free to contact me, I'll happily update the article.

- Plugins can be written in C, Python 3, Scheme, JavaScript and Vala. I will focus on Python 3.
- As with GIMP 2.10, GIMP 3.0 ships with a Python interpreter embedded and will not use the system Python.
- It currently ships Python 3.11
- The C-API documentation is available at <https://developer.gimp.org/api/3.0/>
- The Python API documentation is not yet available to read online, but can be found in a tarball in on the download page: <https://download.gimp.org/pub/gimp/v3.0/api-docs/> It is however of limited benefit, since it doesn't contain any information not in the C-API documentation, except for the exact Python-method names, that can be guessed easily from the names for their C counterparts.
- A basic tutorial on how to write a plugin in Python 3 can be found at <https://testing.docs.gimp.org/3.0/>

[en/gimp-using-python-plug-in-tutorial.html](#)

- Your plugin should be placed in a subfolder of `~/.config/GIMP/3.0/plugin-ins/` and the main file must be named the same as the subfolder.
- GIMP contains a Python REPL that can be started with *Filters -> Development -> Python-Fu -> Console* or quicker with the search feature (/ and then type Python)
- The Python REPL has some code limited completion that sadly only works for classes not for instances.
- You don't need to restart GIMP, when you edit your plugin. It's loaded from disk on every run.
- Run GIMP from terminal to see errors and stdout output of your plugin.
- The GIMP source code repository contains several plugins that can be used as examples. They are located in the [plug-ins folder](#).
- If you want to use any Python modules, other than the standard library, you have to bundle them with your plugin. You can install them in the plugin folder with `pip install --target /path/to/plugin somepackage`. You can use the `site` module to add additional paths to the Python search path.
- If you got stuck, you can ask for help on the IRC channel `#gimp-dev` on `irc.gimp.org`.

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